

UTKARSH GUPTA

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SUMMARY STATEMENT

I am a privacy and security researcher interested in bringing rigorous, theory-grounded methods to real-world deployments of machine learning systems. My work focuses on (i) translating security/privacy objectives into precise mathematical constraints, (ii) formalizing the trade-off between leakage and utility, and (iii) executing adversarial attacks to understand how defense mechanisms fail in practice. I specialize in designing systems that maintain formal security guarantees under strict operational constraints.

EDUCATION

Northeastern University, Boston, USA PhD Candidate in Electrical Engineering Dissertation: Statistical Security Guarantees via Harmonic Analysis: Theory and Applications	Jan 2024 – Present
University of Michigan, Ann Arbor, USA PhD in Electrical Engineering (transferred to Northeastern with advisor); M.S. awarded en route	Aug 2021 – Dec 2023
Indian Institute of Technology Delhi, India B. Tech in Mathematics and Computing (CGPA: 9.46/10)	Aug 2017 – Jul 2021

MAJOR AWARDS AND ACHIEVEMENTS

- Graduated from IIT Delhi in 2021, ranked 2nd in the Mathematics & Computing department.
- Joint Entrance Exam (JEE Advanced) 2017: achieved All India Rank 275 among 1.2 million students.
- Asia-Pacific Mathematical Olympiad (APMO) 2016: Selected to represent India; received Honorable Mention.
- Indian National Mathematical Olympiad (INMO) 2014: Secured 9th position among the top 50 candidates in India.
- Tournament of Towns 2016: Diploma winner; one of 100 students invited to the ToT Math Conference in Russia.
- Scholarships: Awarded the KVPY fellowship (2016) and the NTSE scholarship (2015) by the Government of India.

PUBLICATIONS & PREPRINTS

- **Utkarsh Gupta**, Hessam MahdaviFar, “On the Security of Linear Secret Sharing with General Noisy Side-Channel Leakage,” *Presented at Eurocrypt 2026. Journal version submitted to IEEE Transactions on Information Theory.*
- **Utkarsh Gupta**, Hessam MahdaviFar, and others, “Physical Layer Security via Secret Sharing over Time Modulated Arrays,” *Presented at IMS 2026. Journal version submitted to IEEE Transactions on Microwave Theory and Techniques.*
- **Utkarsh Gupta**, Hessam MahdaviFar, “Bounds on the Statistical Leakage-Resilience of Shamir’s Secret Sharing,” *2024 IEEE International Symposium on Information Theory (ISIT). IEEE, 2024.*
- **Utkarsh Gupta**, Hessam MahdaviFar, “A New Algebraic Approach for String Reconstruction from Substring Compositions,” *Presented at ISIT 2022. Journal version appeared in IEEE Transactions on Information Theory 2024.*
- **Utkarsh Gupta**, Han Mao Kiah, Alexander Vardy, Hanwen Yao “Polar Codes with Balanced Codewords,” *2020 IEEE International Symposium on Information Theory (ISIT). IEEE, 2020.*

RESEARCH EXPERIENCE

- Privacy-Aware User Selection for Multi-Round Secure Aggregation in Federated Machine Learning** Jan 2026 - Present
 - Designed a user-selection protocol that prevents reconstruction of individual client updates from the aggregates observed by the central server across multiple training rounds, under varying user availability.
 - Improved model convergence over the state-of-the-art while guaranteeing privacy, fairness, and robustness to user dropouts. Validated privacy performance against model inversion attacks on models trained using Fashion-MNIST, CIFAR, and ImageNet.
- Formal Privacy Guarantees for Multi-Round Secure Aggregation in Federated Machine Learning** Jun 2026 - Present
 - Proved quantitative privacy guarantees for secure aggregation by bounding how much information a curious server can infer about individual client updates from model aggregates, across multiple training rounds, against curious servers.
 - Proposed a new privacy metric that translates design of user-participation in secure aggregation to quantitative leakage guarantees.
- Security of Secret-Sharing based Masking Schemes against Side-Channel Attacks** Aug 2022 - Jul 2026
 - Improved bounds for the formal security guarantees of masking schemes, characterizing the trade-off between masking order and security.
 - Established these guarantees under general side-channel leakage models, covering a diverse range of eavesdropping attacks.
- Secure Wireless Communications via Time-Varying Beampatterns over Phased Arrays** Jan 2025 - Jul 2026
 - Designed a secret-sharing-based key-management protocol for resource-limited communication (space, IoT networks), that guarantees almost perfect security against eavesdropping. The protocol is energy efficient, and requires no prior knowledge about the adversary.
 - The protocol encodes the message into secret shares, which are transmitted sequentially over time-varying beampatterns, synthesized by a time-modulated array. Security performance is validated using simulations and measurements from a Keysight CATR.

TECHNICAL SKILLS

Security & Privacy: ML privacy, Adversarial ML, Side-channel analysis, Cryptanalysis, Post-quantum cryptography, Secret-sharing
AI & Machine Learning: Federated Learning, Secure Aggregation, Differential privacy, Adversarial robustness, Verifiable ML
Languages: Python, C, C++, Java, MATLAB, SQL, $\LaTeX$
Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Git

OTHER PROJECTS IN SECURITY & PRIVACY

Novel Source Inference Attacks on FedML Systems

May 2026 - Jul 2026

- Designed a novel source-inference attack against state-of-the-art shuffling-based federated-learning defense by adapting methods in membership-inference attacks. Validated its performance for models trained over imbalanced datasets (generated using MNIST, CIFAR).

Adversarial Attacks on Deep-Learning-Based Channel Estimation

Oct 2025 - Dec 2025

- Implemented and adapted gradient-based adversarial attacks for DL-based channel estimation, and designed novel physically realizable over-the-air attacks under practical wireless-system constraints.
- Evaluated adversarial training and defensive distillation as countermeasures, finding that a published defensive-distillation implementation failed to reproduce prior robustness claims and performed worse than the undefended model.

Cryptanalysis of Code-Based Post-Quantum Crypto

Apr 2025 - Jun 2025

- Built and ran a full Information Set Decoding suite (Prange, Stern, MMT, BJMM) against McEliece challenges in a cloud environment, measuring how attack cost scales across the algorithms.
- Prototyped a new repair step for BJMM, based on Kempe chains from graph coloring, that fixes near-miss error vectors in place instead of discarding them, to cut the number of outer iterations.

Microarchitectural Side-Channel Attacks

Sep 2024 - Nov 2024

- Implemented a Meltdown attack to read kernel memory via out-of-order execution, using signal handlers to suppress segmentation faults.
- Implemented a Flush+Reload cache attack to recover OpenSSL's AES key via monitoring cache-hits in T-table usage.

ENTREPRENEURIAL EXPERIENCE

Sanjay – Autonomous Border-Surveillance Tower

Jan 2024 - Apr 2024

System-design proposal submitted to the Indian Army's iDEX program

- Co-developed the end-to-end system architecture for an autonomous AI surveillance tower with multi-modal sensor fusion (optical, thermal, seismic, radar), edge-based CNN/LSTM detection and tracking, and a networked mesh of towers for cross-validated detection.
- Designed the fault-tolerant communication stack: LTE/SATCOM/mesh with dynamic link-switching and network-denied operation, AES-256 with certificate-based authentication, an isolated cryptographic subsystem, and a migration path to post-quantum standards.

Founder and CTO, Credia

Jul 2019 - Jul 2021

Fintech start-up simplifying everyday transactions and credit-access based on transaction history

- Led design of user-behavior driven algorithms to establish a product backend architecture; managed product development from scratch.
- Strategised and developed the deployment of MVP; successfully conducted extensive user surveys to design product flow.

Project Associate, Arth

Dec 2017 - Dec 2018

Social start-up aimed at reducing air pollution in Delhi by replacing wood used in Hindu cremations with cow-dung logs

- Established a sustainable supply chain reducing pollution, urban waste, and empowering small dairy farmers with new sources of income.

RELEVANT COURSEWORK

Security, Cryptography, & Machine Learning:

Computer Hardware & System Security, Verifiable Machine Learning, Machine Learning

Wireless Communications:

Information Theory, Coding Theory, Digital Communications, Advances in Wireless Communications

Mathematical & Algorithmic Foundations:

Probability & Random Processes, Linear Algebra, Approximation Algorithms, Analysis & Design of Algorithms, Data Structures & Algorithms, Combinatorics, Graph Theory, Algebra, Commutative Algebra, Real Analysis, Functional Analysis

TEACHING EXPERIENCE

Teaching Assistant, Northeastern University

Jan 2026 - Apr 2026

- Responsible for grading exams and problem-sets, taking office hours for a probability course for EECE undergraduate students.

Teaching Assistant, IIT Delhi

Jul 2020 - Apr 2021

- Responsible for grading exams, designing problem-sets and conducting tutorial classes for a course in calculus, discrete mathematics.

Instructor, INMO Training Camp

Dec 2017, 2018

- Presented lectures to RMO Delhi region qualifiers at the INMO Training camp RMO regional coordinator.